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Infusion set

Abstract

An infusion set, including a set having a skin contacting surface, a catheter extending from the set, and a flexible ring having an integral fluid path therethrough. The flexible ring is configured to expand to a first diameter to allow placement over the set, and contract to a second diameter to securely circumscribe an outer diameter of the set. The infusion set also includes a tubing length secured to the flexible ring, and a fitting or spike disposed at an end of the tubing length and extending into the flexible ring. The fitting or spike is configured to sealably mate with an opening of the set to permit communication of a medicament therebetween.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 13/138,209, filed on Feb. 1, 2012, which is the U.S. national stage of international patent application No. PCT/US10/00145, filed on Jan. 21, 2010, which claims the benefit under 35 U.S.C. § 119(a) of U.S. Provisional Application No. 61/202,019, entitled “Infusion Set”, filed on Jan. 21, 2009, the entire content, disclosure and subject matter of said application being expressly incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates generally to components, elements and packaging of infusion sets, including features and elements in the areas of tube management, site management, set adhesion, set insertion, set placement and changing operations and packaging.
BACKGROUND OF THE INVENTION
(2) A large number of people, such as those suffering from conditions such as diabetes use some form of infusion therapy, such as daily insulin infusions to maintain close control of their glucose levels. Currently, in the insulin infusion treatment example, there are two principal modes of daily insulin therapy. The first mode includes syringes and insulin pens. These devices are simple to use and are relatively low in cost, but they require a needle stick at each injection, typically three to four times per day. The second mode includes infusion pump therapy, which entails the purchase of an insulin pump that lasts for about three years. The initial cost of the pump can be significant, but from a user perspective, the overwhelming majority of patients who have used pumps prefer to remain with pumps for the rest of their lives. This is because infusion pumps, although more complex than syringes and pens, offer the advantages of continuous infusion of insulin, precision dosing and programmable delivery schedules. This results in closer blood glucose control and an improved feeling of wellness.
(3) As patients on oral agents eventually move to insulin and their interest in intensive therapy increases, users typically look to these insulin pumps for improvements in the management of their condition. Therefore interest in better pump-related therapy is on the rise. In this and similar examples, what is needed to fully meet this increased interest are advanced, improved, and novel new components, elements and packaging of current and future insulin infusion sets, including features and elements in the areas of tube management, site management, set adhesion, set insertion, set placement and changing operations and packaging.
(4) Accordingly, a need exists for such advanced, improved, and novel new components, elements and packaging of current and future infusion sets, that further provide simplicity in manufacture and use improvements for both insulin and non-insulin applications.
SUMMARY OF THE INVENTION
(5) An object of the present invention is to substantially address the above and other concerns, and provide advanced, improved, and novel new components, elements and packaging of current and

future infusion sets, that further provide simplicity in manufacture and use improvements for both insulin and non-insulin applications.

(6) Another object of the present invention is to provide a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user.

(7) Another object of the present invention is to provide an exemplary pushbutton-type inserter, squeeze-type inserter, contact-type inserter, skin pinching-type inserter, folding retraction-type inserter, and/or multistage-type inserter having at least one reusable stage, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(8) Another object of the present invention is to provide an exemplary adhesion means with two or more user-selectable degrees of adhesion strength which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(9) Another object of the present invention is to provide an exemplary self-sealing tube connection means which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(10) Another object of the present invention is to provide an exemplary set having one or more clear and/or magnifying lens features to view a site beneath the set which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(11) Another object of the present invention is to provide an exemplary tube management element having a spring-loaded circular tube reel, tubing pull ties, elastic accordion, pouch or shortened length to manage a tube or tube loop, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(12) Another object of the present invention is to provide an exemplary tube connection element including a tapered connector, detent connector and/or a magnetic attraction connector, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(13) Another object of the present invention is to provide an exemplary adhesion concealment means for concealment of the set once in position and/or a decoration means for decoration of the set once in position, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(14) Another object of the present invention is to provide an exemplary set placement guidance ring and/or one or more finger loops on the inserter to aid in set placement, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(15) Another object of the present invention is to provide an exemplary site preparation wipe or spray which can be provided as part of the inserter, or otherwise included in the single package to simplify assembly and use of the infusion set by the user.

(16) Another object of the present invention is to provide an exemplary package which can hold a number of sets that can be easily released and retrieved from the tray by an inserter to simplify assembly and use of the infusion set by the user.

(17) Another object of the present invention is to provide an exemplary insertion needle handle and shroud which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(18) Another object of the present invention is to provide an exemplary squeeze-type latch between an upper portion and a lower portion of the set, and/or a tool removable upper portion of the set, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

(19) Another object of the present invention is to provide an annular fluid reservoir and/or fluid path in the set hub, which can significantly minimize the penetration distance of the tubeset connector into the hub while maintaining a sufficiently large tubeset connector geometry.

(20) Another object of the present invention is to provide substantially closed or sealed annular fluid reservoir and/or fluid path in the set hub, which can reseal any insertion openings generated

by the insertion needle, and allow penetration by a tubeset connector needle.

(21) Another object of the present invention is to provide an infusion set constructed of a soft, pliable and/or elastic or similar material such that the infusion set is soft or pliable to a degree that allows the elasticity of the materials to affix a tube "ring" of the tubeset connector to the hub.

(22) Another object of the present invention is to provide an infusion set constructed such that the tube ring of the tubeset connector can include the tubeset connector needle to pierce the hub, wherein the elasticity of the materials function to seal the insertion site of the tubeset connector needle.

(23) Another object of the present invention is to provide an infusion set constructed such that the tubeset connector needle of the tube ring of the tubeset connector can pierce the hub at any rotational position, and wherein the elasticity of the materials function to seal the insertion site of the tubeset connector needle such that the tubeset connector needle of the tube ring of the tubeset connector can be withdrawn and the tube ring repositioned at a different rotational alignment position.

(24) Another object of the present invention is to provide a catheter constructed of a body temperature softening polyurethane or similar material, and include one or more features including a splined lumen and holes or openings along a body length, including cross-drilled holes.

(25) These and other objects are substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more of an exemplary pushbutton-type inserter, squeeze-type inserter, contact-type inserter, skin pinching-type inserter, folding retraction-type inserter, and/or multistage-type inserter having at least one reusable stage. One or more of the exemplary embodiments comprise a user gripping surface and a means to release a firing spring for set placement, and one or more retraction features to retract the insertion needle to avoid any dangers to the user and permit safe disposal. Further, one or more of the exemplary embodiments can comprise a set placement guidance ring to be placed on the insertion site and which mates with an end of the inserter to ensure that the inserter is properly positioned before release of the set. Further, one or more of the exemplary embodiments can comprise one or more finger loops extending from a body of the inserter to aid in set placement. Still further, one or more of the exemplary embodiments can comprise a squeeze-type latch between an upper portion and a lower portion of the set comprising one or more pushbuttons which articulate a latch between the upper and lower portions, thereby releasing the upper portion of the set from the lower portion which can remain in position. In yet other embodiments of the present invention, the upper portion can be configured to be tool-removable, such that the tool comprises one or more pins which are inserted into the lower portion and release a latch between the upper and lower portions, thereby releasing the upper portion of the set from the lower portion which can remain in position. Still further, one or more of the exemplary embodiments can comprise an insertion needle handle and shroud comprising one or more hinged flat members which can be bent to cover the protruding needle after use.

(26) These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more of an exemplary adhesion means with two or more user-selectable degrees of adhesion strength comprising at lease a first and second adhesion ring having different degrees of adhesive strength. The user can select which ring to use by removing a cover of the desired ring and leaving the remaining rings covered. A self-sealing tube connection means can be provided and comprise a set having a groove into which an elastic ring and tube connector can be positioned, such that the elastic ring secures the tube to the set and seals the connection between each. The set can comprise one or more clear and/or magnifying plastic components to view a site beneath the set, wherein the adhesive pad is provided with one or more clearances to allow the visual access.

(27) These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more tube management elements comprising a spring-loaded circular tube reel which serves to feed and retract excess tube as urged by a reel spring, one or more tubing pull ties which can be pulled to either secure or release tubing being stored in a large loop, an elastic accordion or pouch in which to store tubing, or provide shortened tube lengths between set and pump. Such tubing can further comprise one or more embodiments of a tube connection including a tapered connector in which a tight engagement provides connection, a detent connector in which projecting detents and recessed detents provide connection, and a magnetic connector in which magnetic attraction provides connection. Still further, an insulin supply comprising an insulin content, supply vial and tubing can be provided with the tubing and tubing connectors.
(28) These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more of an adhesion concealment means for concealment of the set once in position comprising an adhesive pad to be placed over a set to conceal the set, or a decorative adhesive pad to be placed over a set to enhance the set. A site preparation wipe comprising a disposable pad including a site preparation solution can be provided as well as an improved inserter body that comprises a spray mechanism and site preparation solution contents.
(29) These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in an exemplary package comprising a number of openings into which sets can be aligned and secured by a covering, such as foil, and which allows set removal by an inserter configured to pierce the foil and capture and remove the set from the tray.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The various objects, advantages and novel features of the preferred embodiments of the present invention will be more readily appreciated from the following detailed description when read in conjunction with the appended drawings, in which:
(2) FIGS. **1** and **2** are perspective views of a collection of infusion set elements and associated packaging in accordance with an exemplary embodiment of the present invention;
(3) FIGS. **3***a*-**3***e* are cross-sectional views of an exemplary single-button insertion device in accordance with an exemplary embodiment of the present invention;
(4) FIGS. **4***a*-**4***c* are perspective views of exemplary set adhesion elements in accordance with an exemplary embodiment of the present invention;
(5) FIGS. **5***a*-**5***d* are views of an exemplary set connection method in accordance with an exemplary embodiment of the present invention;
(6) FIGS. **6***a*-**6***b* are views of an exemplary set site inspection element in accordance with an exemplary embodiment of the present invention;
(7) FIGS. **7***a*-**7***b* are views of an exemplary set tubing management element in accordance with an exemplary embodiment of the present invention;
(8) FIG. **8** is a perspective view of an alternate insulin supply and associated tubing in accordance with an exemplary embodiment of the present invention;
(9) FIGS. **9***a*-**9***b* are perspective views of an exemplary set concealment element in accordance with an exemplary embodiment of the present invention;
(10) FIGS. **10***a*-**10***c* are perspective views of exemplary set placement elements in accordance with

an exemplary embodiment of the present invention;
(11) FIG. **11** is a perspective view of an exemplary set site placement preparation element in accordance with an exemplary embodiment of the present invention;
(12) FIGS. **12***a*-**12***f* are views of exemplary sets on tray packaging in accordance with an exemplary embodiment of the present invention;
(13) FIGS. **13***a*-**13***f* are views of an exemplary multistage insertion device in accordance with an exemplary embodiment of the present invention;
(14) FIGS. **14***a*-**14***d* are cross-sectional views of an exemplary "squeeze-type" inserter in accordance with an exemplary embodiment of the present invention;
(15) FIGS. **15***a*-**15***e* are views of an exemplary "contact-type" inserter in accordance with an exemplary embodiment of the present invention;
(16) FIGS. **16***a*-**16***c* are views of an exemplary needle handle and shroud in accordance with an exemplary embodiment of the present invention;
(17) FIGS. **17***a*-**17***d* are views of an exemplary "skin pinch-type" inserter in accordance with an exemplary embodiment of the present invention;
(18) FIGS. **18***a*-**18***e* are views of an exemplary "folding/retracting-type" inserter in accordance with an exemplary embodiment of the present invention;
(19) FIGS. **19***a*-**19***c* are views of an exemplary "squeeze-type" latch provided as a connection method in accordance with an exemplary embodiment of the present invention;
(20) FIGS. **20***a*-**20***d* are views of an exemplary "tool-type" latch provided as a connection method in accordance with an exemplary embodiment of the present invention;
(21) FIGS. **21***a*-**21***c* are perspective views of an exemplary pull tie, tubing management element in accordance with an exemplary embodiment of the present invention;
(22) FIGS. **22***a*-**22***c* are perspective views of exemplary tubing connecting elements in accordance with an exemplary embodiment of the present invention;
(23) FIGS. **23***a*-**23***b* are perspective views of an exemplary elastic accordion, tubing management element in accordance with an exemplary embodiment of the present invention;
(24) FIGS. **24***a*-**24***c* are perspective views of an exemplary short tube, tubing management element in accordance with an exemplary embodiment of the present invention;
(25) FIGS. **25***a*-**25***b* are perspective views of exemplary pouch-type, tubing management elements in accordance with an exemplary embodiment of the present invention;
(26) FIGS. **26***a*-**26***c* are perspective views of exemplary decorative, functional covering elements in accordance with an exemplary embodiment of the present invention;
(27) FIGS. **27***a*-**27***b* are perspective views of an exemplary two-finger, placement element in accordance with an exemplary embodiment of the present invention;
(28) FIG. **28** is a perspective view of an exemplary insertion tool-mounted spray, site preparation element in accordance with an exemplary embodiment of the present invention;
(29) FIG. **29***a*-**29***c* are views of an exemplary hub including an annular fluid reservoir and/or fluid path for use with one or more exemplary embodiments of the present invention; and
(30) FIG. **30** is a cross-sectional view of a catheter including one or more cross-drilled holes and a splined lumen in accordance with an exemplary embodiment of the present invention.
(31) Throughout the drawings, like reference numerals will be understood to refer to like parts, components and structures.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(32) The embodiments of the present device described below illustrate a number of advanced, improved, and novel new components, elements and packaging of current and future infusion sets, that further provide simplicity in manufacture and use improvements for both insulin and non-insulin applications. Exemplary embodiments are presented in separate descriptions, although the individual features of these embodiments can be combined in any number of ways to meet the needs of the user.

(33) As will be appreciated by one skilled in the art, there are numerous ways of carrying out the examples, improvements and arrangements of insulin-associated devices disclosed herein. Although reference will be made to the embodiments depicted in the drawings and the following descriptions, the embodiments disclosed herein are not meant to be exhaustive of the various alternative designs and embodiments that are encompassed by the disclosed invention.
(34) The embodiments of the present device described below illustrate a number of features and elements in the areas of tube management, site management, set adhesion, set insertion, set placement and changing operations and packaging. A collection of exemplary embodiments of the present invention is shown by way of example in FIGS. **1** and **2**, which serve to introduce elements described in greater detail below. FIG. **1** illustrates an exemplary infusion set **10** including the following features. As shown in FIG. **1**, an exemplary infusion set can comprise an inserter, such as the squeeze-type inserter **700** for use with a set, such as the ring-sealed set **350**. A tube and associated tube management devices, such as the circular reel **450**, can be provided for communication with an insulin pump (not shown) or with an insulin supply, such as the insulin supply **475**. A placement assistance element can be provided, such as the placement assistance ring **526**, and the entire arrangement can be placed into a sealed tray **12** with a number of site preparation elements, such as the pads **550**, and site concealment elements, such as the pads **500**. The tray can be comprised of any suitable plastic, fiber or composite material compatible with the components, and can provide compartments, padding or element securing detents or moldings. The set can be packaged in the tray **12** and sealed with a clear and/or labeled cover **14**, and includes every component needed to position, connect, insert, and wear the set, as well as the insulin **475** itself as shown in FIG. **2**. Each of the exemplary components, including a number of additional or alternate components, will now be described individually in greater detail.
(35) An exemplary embodiment of the present invention can be provided with an insertion device as desired by a user. An exemplary insertion device **100** is shown in FIGS. **3***a*-**3***e*. The exemplary insertion device of FIGS. **3***a*-**3***e* provides an insertion device which can contain the set at an open, patient-contacting end, and provide an actuation button at an opposite end. Upon activation, the insertion device places the set and automatically retracts the insertion needle back into the insertion device.
(36) As shown in FIG. **3***a*, the insertion device can comprise a substantially cylindrical housing **102** from which a spring biased projection or pushbutton **104** can extend, and in which a set **106** can be positioned for use. The housing **102** can have a first diameter at a lower portion thereof in which the set **106** can be positioned, and transition to a reduced second diameter at an opposite end to substantially equal a diameter of the pushbutton **104**.
(37) Within the body of the housing **102**, a number of elements are contained which serve to fire the set **106** into position, and then retract the insertion needle. To do so, the housing **102** comprises a first, second and third chamber **108**, **110** and **112** of different widths. The first chamber **108** at an uppermost portion of the housing **102** has a width sufficient to slidably receive an end of an inserter rod **114**. At an upper end of the first chamber **108**, an opening is provided though which the pushbutton **104** slidably enters the first chamber **108**, and a lower end of the first chamber opens to create the second chamber **110** as described in greater detail below. The first chamber further comprises at least one inclined detent **116** disposed upon an inner wall of the first chamber **108** which serves to capture and secure a similar, deflectable detent **118** at an upper end of the inserter rod **114**. One or more further similar, deflectable detents **120** are disposed upon an end **122** of the pushbutton **104** within the first chamber **108**. The end **122** of the pushbutton **104** is provided to have a width substantially equal to the width of the first chamber **108** to align and guide the pushbutton **104** during operation. A spring **124** is disposed concentrically with the pushbutton **104** and is captured between an outer surface of the housing **102** and an expanded head of the pushbutton **104** to constantly urge the pushbutton **104** upward.
(38) As noted above, the pushbutton **104** comprises an end **122** with one or more inclined detents

120. When pressed downward, the inclined detents **120** come into contact with the inclined detents **118** of the inserter rod **114** which are being held by the detents **116** of the first chamber **108**. The contact releases the detents **118** and allows the inserter rod **114** to be urged downward by a firing spring **126** disposed within the second chamber **110** of the housing **102**.

(39) The second chamber **110** has a width wider than that of the first chamber **108**, which is sufficient to slidably receive the inserter rod **114** and more specifically, a width sufficient to slidably contain planar members **128** and **132** of the inserter rod **114**.

(40) The firing spring **126** is disposed concentrically with the inserter rod **114** about an outer circumference of the inserter rod body as captured within the second chamber **110**. That is, the firing spring **126** is captured within the second chamber **110** of the housing **102** between an upper wall of the second chamber **110**, and the upper surface of the planar member **128** of the inserter rod **114**. In doing so, the firing spring **126** constantly urges the inserter rod **114** downward.

(41) At a lower portion of the housing **102**, the third chamber **112** can be provided having a width wider than that of the second chamber **110**, thereby creating a shoulder **130** therebetween. As described in greater detail below, the shoulder **130** is configured to allow downward travel of the inserter rod **114** and capture one or more detents on an outer surface of the inserter rod **114** to prevent retraction of the inserter rod **114**, yet permit full retraction of a needle carrier and inserter needle.

(42) As noted above, the inserter rod **114** is configured to slidably travel though each of the first, second and third chambers **108**, **110**, and **112**, of the housing **102**. The inserter rod **114** is substantially cylindrical and comprises a diameter at an upper portion substantially equal to the width of the first chamber **108** to be directed and guided by the first chamber **108** during use. A lower portion of the inserter rod **114** comprises the first and second planar members **128** and **132**, which comprise a diameter substantially equal to the width of the second chamber **110** to be directed and guided by the second chamber during use.

(43) As noted above, the inserter rod **114** comprises the first and second planar members **128** and **132**. The inserter rod **114** further comprises an inserter needle guide **144** and at least third and fourth elements **136** and **138** extending between the first and second planar members **128** and **132**. A spring **140** is captured between the needle guide **144** and fourth element **138**, and passes through an opening in the third element **136**, to constantly urge both the third and fourth elements **136** and **138** outward from a center axis of the insertion device, and against an inner wall of the second and third chambers **110** and **112**. The fourth element **138** comprises an inclined detent **142** which is urged against the inner wall of the second and third chambers **110** and **112**. In doing so, as the inserter rod **114** is urged downward and the detent **142** passes the shoulder **130**, the spring **140** urges the element **138** against the inner wall of the third chamber **112** such that the detent **142** is captured by the shoulder **130** and prevents the retraction of the inserter rod **114** as shown in FIGS. **3***c* and **3***d*. The needle carrier and inserter needle are permitted to retract as described in greater detail below.

(44) The third element **136** comprises at least one projection **146** which is configured to capture a groove **154** in the needle carrier **148** slidably disposed within an inner opening of the inserter rod **114**. The needle carrier **148** secures the inserter needle **134** at a lower end, such that the inserter needle extends through the needle guide **144** and though an opening in the planar member **132** of the inserter rod **114**. The needle carrier **148** further comprises the groove **154** which can be captured by the projection **146**. The remainder of the needle carrier **148** extends through the planar member **128** and into the inner opening of the inserter rod **114** and terminates at a planar end **150**. The planar end **150** of the needle carrier **148** has a width substantially the same as the width of the inner opening of the inserter rod **114** to align and guide the needle carrier **148** during retraction. A retraction spring **152** is captured between the planar end **150** of the needle carrier **148** and the planar member **128** of the inserter rod **114**. In doing so, the retraction spring **152** constantly urges the needle carrier **148** upward.

(45) While in the pre-use position, a large portion of the second chamber **110** remains open to the end of the device. In doing so, the set **106** can be positioned on the extended needle **134**, at an opposite side of the second planar member **132** of the inserter rod **114**. The set **106** can be gently held within the second chamber **110** through contact with the walls of the chamber, and/or through contact with the inserter needle **134**. As noted elsewhere, the set **106** can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the insertion device **100**.
(46) As shown in FIGS. **3***a*-**3***e*, the compression of the push button **104** releases the inclined detents **118** of the inserter rod **114** as shown in FIG. **3***a*, permitting the firing spring **126** to drive the needle **134**, set **106**, and adhesive pad into the region of the skin beneath the third chamber **112** of the device as shown in FIG. **3***b*, and also releases the spring **140** to lock the inserter rod **114** in the extended position as shown in FIG. **3***c*. That is, upon release, the inserter rod **114** is free to travel downward as urged by the trapped spring **126**. In doing so, the inserter rod **114**, including its end **132** and needle **134** travel downward through the third chamber **112**, urging the set **106** downward with it. At or before reaching the travel limit of the inserter rod **114**, the set **106** is positioned, retraction of the needle carrier **148** and needle **134** occurs, and the device can be removed as shown in FIG. **3***d*, thereby leaving the set **106** at the desired insertion site as shown in FIG. **3***e*.
(47) To retract the needle carrier **148** and needle **134**, no user action is required. As shown in FIGS. **3***c*-**3***d*, the shoulder **130** serves to hold the inserter rod **114** in the down position. When the insertion is completed and the inserter rod **114** is captured in the down position by the outward movement of members **136** and **138**, the movement simultaneously releases the needle carrier **148** and needle **134**, and allows the needle carrier **148** and needle **134** to retract upward as urged by the retraction spring **152** as shown in FIGS. **3***c* and **3***d*. The needle carrier **148** and needle **134** are retracted until contacting an upper stop **156** disposed at an upper end of the inserter rod **114**.
(48) The insertion device **100** of FIGS. **3***a*-**3***e* uses the single button press of the pushbutton **104** to release the firing mechanism, insert the set **106**, and safely retract the needle **134** after insertion. The insertion device **100** is packaged and stored with the firing mechanism in an upright and locked position, with the firing spring **126** compressed as shown. The inserter rod **114** has the flexible wedge-shaped tabs or latches **120** at the top of it, which are locked into grooves or captured by the detents **116** at the interior top of the inserter's body **102** as shown in FIG. **3***a*.
(49) Pressing the activation button **104** unlatches the firing tabs at the top of the unit as shown in FIG. **3***b*, and permits the firing spring **126** to extend toward its free length, propelling the needle carrier assembly **148** of the inserter rod **114** downward as shown in FIG. **3***c*. Upon reaching the wider region of the body interior shaft at the third chamber **112**, the two spring-loaded locking halves or members **136** and **138** of the needle carrier assembly **148** move outward and lock the firing mechanism in the down position as shown in FIG. **3***d*, and frees the central portion of the needle carrier assembly **148** to move upward as shown in FIG. **3***e*, leaving the remainder of the inserter rod **114** in the down position.
(50) By this time, the needle **134** has pierced the skin and placed the cannula and set **106**, adhering it to the skin. Having been decoupled from the inserter rod **114** which has been driven downward and locked, the needle carrier assembly **148** is now free to rebound, propelled upward by the retraction spring **152** as shown in FIG. **3***e*. After retracting fully upward, the needle **134** is stored permanently and inaccessibly inside the insertion device housing **102**, and the device is inert.
(51) In this and other inserter embodiments described below, the inserter body and elements can be constructed of any suitable and compatible materials such as plastic or metal. Springs can be provided as coil springs made of plastic or metal, although embodiments of the present invention are not limited thereto, and other spring or biasing means can be used, such as leaf spring or simply material resiliency. The insertion needle can comprise any suitable set insertion needle of metal or plastic, having length, thickness, and bevel dimensions suitable for set insertion.
(52) One or more of the exemplary embodiments of the present invention described herein can be

further provided with additional features or elements to manage the degree of adhesion of the device in some manner as desired by a user. As shown by way of example, adhesion management elements **300** are shown in FIGS. **4***a*-**4***c*. As shown in FIGS. **4***a*-**4***c*, a set **314** and catheter **316** are shown encircled by concentric rings of adhesive **302** and **306**. In an exemplary embodiment of the present invention, one ring can provide an adhesive with a higher degree of adhesion strength, and another ring can provide an adhesive with a lesser degree of adhesion strength, thereby allowing a user to tailor the degree of set adhesion to the user's activity plans. Although only two concentric rings of varied adhesive are shown in FIGS. **4***a*-**4***c*, in yet other embodiments of the present invention, more rings can be provided, or the rings may be provided in alternate, non-circular shapes (i.e., such as oval shapes). Further, in the exemplary embodiment shown in FIGS. **4***a*-**4***c*, the inner ring **306** is provided with the adhesive with a higher degree of adhesion strength and the outer ring **302** is provided with the adhesive with a lesser degree of adhesion strength, but embodiments of the present invention are not limited thereto. In yet other embodiments of the present invention, the order can be reversed or additional rings of adhesive provided.

(53) As noted by those skilled in the art, a set typically requires a degree of adhesion to maintain proper positioning of the device. Accordingly, the exemplary embodiments of the present invention provide a set that comprises an adhesive pad or patch on the underside with a user-configurable adhesion mechanism to adapt the set to the expected environment in which it will be worn. As shown in FIG. **4***a*, the adhesive pad or patch can be provided in two or more, selectable strengths which can be variably exposed and utilized depending on the user's preference. A region of basic adhesive **302** having a removable cover **304** segmented from other covers, can be used for everyday needs while the region of extra-strong adhesive **306** remains covered by a similar segmented cover **308** as shown in FIG. **4***b*. The region of extra-strong adhesive **306** can be exposed by the removal of the cover **308** if athletic activities might be expected to stress the set adhesion as shown in FIG. **4***c*. To further simplify use, each cover **304** and **308** of each region can comprise a tab **310** and **312**, respectively, to aid in removal of each adhesive cover. The covers **304** and **308** can also be labeled, color-coded or textured to show the adhesive strength thereunder.

(54) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective connection system as desired by a user. As shown by way of example, such a connection method **350** is shown in FIGS. **5***a*-**5***d*.

(55) In such an exemplary embodiment, a set **352**, once placed, has a port **354** that should be able to be easily connected and disconnected with a tubing **358** leading to an insulin pump (**359**). To do so, the set **352** can comprise the "self-sealing" connection port **354** on the outer, circular perimeter of the set **352**, within a circumferential groove **356** on the body of the set **352** as shown in FIG. **5***a*. An incoming tube **358** can comprise a fitting **360** at the end designed to securely fit into and seal with the port **354** on the set **352**, using the assistance of a flexible, resilient ring **362** extending outward from it as shown in FIG. **5***b*.

(56) Connection of the tube **358** to the set **352** can then be accomplished by stretching the ring **362** around the far side of the set **352** as guided by the groove **356** to a first diameter to allow placement, then placing the connection fitting **360** into the port **354** as shown, and allowing the elastic ring **362** to contract to a second diameter to retain it securely as shown in FIGS. **5***c* and **5***d*. The tube, set and connectors can be constructed of any suitable material as described herein, and the ring **362** can be constructed of any compatible, resilient material which can be easily molded into the desired shape and maintain elasticity at least for an expected shelf life of the device. In a similar manner, the exemplary embodiment of the present invention shown in FIGS. **29***a*-**29***c*, described in greater detail below, comprises an infusion set constructed of a soft, pliable and/or elastic or similar material such that the infusion set is soft or pliable to a degree that allows the elasticity of the materials to affix the tube ring of the tubeset connector to the hub in any number of rotational positions and which further includes a tubeset connector needle to pierce the hub,

wherein the elasticity of the materials function to seal the insertion site of the tubeset connector needle.
(57) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to allow site inspection in some manner as desired by a user. As shown by way of example, a site inspection embodiment 400 is shown in FIGS. **6***a*-**6***b*. As shown in FIGS. **6***a*-**6***b*, a set **404** and its housing can include a means to inspect the region of the skin immediately surrounding the insertion point, to ensure that the site is in good condition. In an exemplary embodiment of the present invention, the set **404** can include an element **402** extending from a top surface to a bottom surface of the set **404** at some point near the insertion site. As shown in FIGS. **6***a*-**6***b*, the element **402** completely encircles the site, but embodiments of the present invention are not limited thereto. In yet other embodiments of the present invention, the element **402** can be provided over a narrower portion, but still sufficient to view the site from above the device.
(58) The element **402** can be constructed of any suitable material that can be easily manufactured, bonded with the remaining elements of the set **404**, provide compatibility with the contents or other materials, including the skin surface, and provide a degree of visibility between the top and bottom surfaces of the set **404**. As shown in FIG. **6***b*, the sides of the element **402** can be configured, contoured or otherwise provided with features to be held in place by the body of the set **404**, and a top surface can be contoured to add a degree of magnification. For example, the housing of the set **404** or the top surface of the element **402** can include or comprise a clear plastic magnifying element that allows for even better site inspection abilities.
(59) Further, the element **402** can be provided with an upper surface opening **412** to secure a septum **414** and for insertion of a placement needle **408** into a catheter **410**. The element **402** can further provide an opening **416** to facilitate introduction of the infusion substance through the tubing **406**. Still further, where the lower surface of the set **404** is provided with an adhesive pad (not shown) at the base of the set **404**, the adhesive pad can include a cutaway portion to permit visibility through the element **402**.
(60) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to secure, contain, and/or conceal the tubing of the device in some manner as desired by a user. As shown by way of example, a tube management reel **450** is shown in FIGS. **7***a*-**7***b*. The tubing **452** connecting the insulin supply and pump (not shown) to the infusion set (not shown) can be packaged on a spring-loaded circular reel **454** disposed within or at one side of a reel housing **456**.
(61) As shown in the cross-sectional view of FIG. **7***b*, the tubing **452** can enter and exit the housing **456** at opposite sides near an upper surface, and wrap about a spring mechanism using pins **458**. The construction of a spring-loaded circular reel is known to those skilled in the art, so additional features of which are omitted for clarity. However, the circular reel is provided with pins **458** between which the tube **452** is secured within the housing **456** such that, feeding tubing from the housing results in the circular reel being wound tighter, and feeding tubing into the housing results in the circular reel being unwound and relaxed. Accordingly, the circular reel and pins maintain a tension on the tube **452** urging the tubing into the housing.
(62) The circular reel can further comprise a catch/latch mechanism as known to those skilled in the art such that pulling the tube **452** a first time feeds a length of tube and a catch is provided to prevent a reverse spring-urged action. Upon pulling the tube **452** a second time, the catch can be released so that the reverse spring-urged action is released to urge the tube **452** back into the housing **456**. In doing so, the reel device allows slack tubing to be fed out precisely, with spring resistance maintaining the excess tubing rolled up and stored. The locking switch or latch can be provided to allow the user to prevent inadvertent retraction or extension once a satisfactory length of tubing has been deployed. The device can further comprise a switch **460** to actuate the tube retrieval.

(63) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide an insulin supply in some manner as desired by a user. As shown by way of example, an insulin supply **475** is shown in FIG. **8**. FIG. **8** illustrates an exemplary embodiment of an insulin container **476** and associated tubing **478**, including a tubing connection means **480**. The insulin supply **476** can be packaged in a small, sealed or sealable container that is pre-connected to a length of tubing **478**. The insulin container **476** can be integrated with a pump mechanism (not shown) externally via an installation process which the user can easily perform. The insulin container **476** and tubing **478**, once connected to a set and pump, form a system that does not need to be primed for proper function.
(64) The insulin container **476** can be constructed of any suitable material, such as glass or plastic, to be clear to show the contents, or non-clear or opaque to protect contents from light. The container **476** can further include incremental dosage measurement marks along one or more surfaces for use during content delivery. The associated tubing **478** and connection means **480** can be constructed of any suitable material, such as rubber, to provide flexibility and compatibility with the contents. The connection means **480** can be constructed in any number of ways, for example, including the connection means described in greater detail below in regard to FIGS. **22***a*-**22***c*.
(65) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide set concealment in some manner as desired by a user. As shown by way of example, a concealment element **500** is shown in FIGS. **9***a*-**9***b*. In FIG. **9***a*, an adhesive covering **502**, similar to a large adhesive bandage, can be provided to enable the user to cover the site, including the set **504**, with an inconspicuous dressing. In an exemplary embodiment of the present invention, the covering **502** comprises a flexible, skin-colored adhesive covering that can have an adhesive side and a non-adhesive side. The adhesive side can be covered with a user-removable cover (not shown) that when removed, allows the covering **502** to be secured over the site, thereby covering and to a large degree, concealing the set **504** as shown in FIG. **9***b*.
(66) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to aid and/or simplify placement of the device in some manner as desired by a user. As shown by way of example, a collection of placement assistance elements **525** is shown in FIGS. **10***a*-**10***c*. Since the set and the corresponding insertion device can be unavoidably large in diameter and difficult to accurately locate in some circumstances, an exemplary kit including embodiments of the present invention can further include a placement ring to aid in placement of the set. An exemplary placement ring **526** is shown in FIG. **10***a* and can comprise a ring-shaped plastic part with one or more orienting features **530**, such as keys, on its perimeter, and a self-adhesive, covered pad on the underside (not shown). A low-profile, contoured circular plastic ring is shown, but embodiments of the present invention are not limited thereto. As the ring **526** can be provided with an adhesive pad, the ring **526** can be first gently adhered to the skin surface with the target insertion site at the center as shown in FIG. **10***b*. This allows careful set placement as the insertion site can now be better visualized though a center opening of the ring **526**, and the insertion device **528**, or tool, can be aligned with, and guided into final position, by the placement ring **526**.
(67) The insertion device **528** can be provided for use with the placement ring **526**, and be constructed as described elsewhere herein and further having corresponding detents or keyways **532** to align with and receive the orienting features **530** of the placement ring **526**. When the insertion device **528** is to be placed atop the placement ring **526** as shown in FIG. **10***c*, the insertion device **528** self-aligns and orients for precise location of a set **534**. In the exemplary embodiment shown, the orienting features **530** are formed as raised contoured detents. Therefore, each further serves to guide, center and align the insertion device **528** upon placement. That is, such contoured elements provide a degree of self-alignment not as readily provided by square elements for example. After insertion, the set **534** is left remaining at a center of the ring **526** upon removal of

the insertion device **528**. The placement ring **526** can then be removed and discarded.

(68) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional elements for the use of site preparation. Such an exemplary feature is shown in element **550** of FIG. **11**. In the exemplary embodiment shown in FIG. **11**, set packaging can further include a site preparation wipe **554** contained within a preservation container **552**. The packaging for the set can include the wipe **554**, such as a versatile disposable wipe, paper or cloth pad that is soaked or impregnated with one or more of a disinfectant, local anesthetic or other helpful substance. The pad or wipe **554** may also be constructed having a texture, coating, or other surface feature **556** that can provide an exfoliating ability to aid in anesthetic effectiveness.

(69) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective means to provide packaging as a number of sets upon a tray as desired by a user. As shown by way of example, a "sets on tray" packaging arrangement **575** is shown in FIGS. **12***a*-**12***f* As shown in FIGS. **12***a*-**12***f*, a number of exemplary sets **576** can be packaged upon a tray **578**. Although the exemplary tray of FIG. **12***a* shows the containment of three sets, any number or arrangements of sets can be included as desired by the user.

(70) As shown in FIG. **12***a*, a number of disposable sets or set components **576** can be packaged in the exemplary multi-unit, foil or plastic-sealed tray **578**. In the exemplary embodiment of the present invention shown in FIG. **12***a*, the tray **578** can be constructed of any suitable material compatible with the set and set components to be stored therein, and can provide a number of recessed, contoured or otherwise constructed openings **580** into which the sets or set elements **576** can be positioned. The openings **580** can be configured to securely hold and protect the sets prior to use, allow easy covering of the sets and tray surface with a sealing means, such as foil or other material which can then be easily removed or punctured by the user to access the desired set and maintain protection of remaining sets as shown in FIG. **12***a*, and which allows such access and removal using an inserter device **582** as shown in FIGS. **12***c*-**12***f*.

(71) As illustrated in the cross-sectional views of FIGS. **12***c*-**12***f*, an exemplary insertion device **582** is shown having a contoured shape into which the set **576** can be captured. To do so, the insertion device **582** can comprise one or mover deflectable ends **584** which can have an inclined latch **590** to pierce the tray covering **588** as shown in FIG. **12***d*, deflect outward slightly due to contact with the set **576**, and then capture an outer circumference of the set **576** once the latch **590** is fully inserted. In doing so, the insertion device **582** can be used to extract a new set **576** from the package so that the user will not need to contend with opening and unsealing packaging materials. For example, an exemplary and reusable insertion device **582** for use which such a tray can comprise a hollow underside or lower surface, with the engagement features of the latches **590** oriented inward towards the set **576**, such that a user is simply required to align the insertion device **582** with a set, insert to a sufficient depth and retrieve the set **576** for use. No further user action is required in regard to handling the set **576**.

(72) As shown in FIG. **12***c*, the sets **576** are aligned within each opening of the tray with sufficient clearance below the set to accommodate any elements of the set. A substantial portion of the upper surface of the set can be exposed, wherein a foil or other covering **588** can be used to secure the set **576** within the tray, and seal the contents of the tray and set **576** from contamination or other damage. In this exemplary embodiment of the present invention, the foil **588** is shown covering the limited space surrounding each device, but is not limited thereto. In yet other embodiments of the present invention the foil or covering can be more or less extensive upon the tray surface as desired.

(73) As shown in FIG. **12***d*, the insertion device **582** can self-align on the blister-type package of the set **576** on the tray **578**, and when pressed down, can cut through the foil or plastic seal **588** as shown in FIGS. **12***c* and **12***d*. As the insertion device **582** presses down further, it also disengages the set **576** from the packaging tray **578** with the perimeter of the inserter body. After pushing past

the edge of the set **576**, the locking tabs or latches **590** on the insertion device **582** engage and secure the outer circumference of the set **576**, and allow the user to extract both the insertion device **582** and the set **576** from the tray **578** as shown in FIG. **12***f*. The insertion device and the set are then ready to fire in normal use. The remaining sets of the tray are left intact and ready for later use. In exemplary embodiments of the present invention, the removal of the set **576** from the tray also results in the automatic removal of any needle cover and adhesive backing, which is left with the tray **578**.

(74) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective multistage-type inserter as desired by a user. As shown by way of example, a "multistage-type" inserter device **600** is shown in FIGS. **13***a*-**13***f*. In such an exemplary embodiment, the insertion device can be constructed in such a way as to separate high-cost parts from low-cost parts, keeping the former in a reusable mechanism while allowing the latter to be safely disposed.

(75) As shown in FIG. **13***a*, the multistage-type inserter device **600** can comprise a substantially cylindrical upper and lower element **602** and **604**, respectively. The upper element **602** can have a first diameter at a lower portion thereof to seamlessly mate with the lower element **604**. An upper portion of the upper element **602** can have a second diameter which is flared or expanded to provide sufficient room for operation of hinged latches **606** as described in greater detail below. The lower element **604** can also have a first diameter at an upper portion thereof to seamlessly mate with the upper element **602**, and a second diameter at a lower portion which is flared or expanded to contain a set **642**.

(76) The upper element **602** can comprise at least one firing mechanism consisting of at lease one hinged latch **606** rotatable about a pin **608** or other means, and which has a inclined projection **610**. Each projection **610** includes an inclined lower surface to facilitate assembly with the lower element **604**, and a substantially flat upper surface to restrict some portion of an inserter rod **614** as described in greater detail below. The upper element **602** further comprises a first chamber **612** in which a firing spring **624** is captured. The upper element **602** and contents thereof can comprise a reusable element that can be installed onto a disposable mechanism of the lower element **604** which can include a set, needle, adhesive pad, and a portion of the insertion mechanism.

(77) The inserter rod **614** extends though both the upper and lower elements **602** and **604**, and comprises a cross-member **616**, a shoulder **618** and a planar end **620**. The inserter rod **614** further comprises an inserter needle **622**, which can be secured within a center opening of the inserter rod **614**, and can extend downward from the rod at the end **620**. Both the cross-member **616** and the lower end **620** are configured to have a width substantially equal to the width of the chamber in which each is positioned to facilitate alignment and travel of the inserter rod **614** during use. As described in greater detail below, the cross-member **616** is configured to be held in an up position by the projections **610** of the latches **606**, and is configured to be blocked at a down position by the projections **633** of the lower element **604**. Further, the lower part of the shoulder **618** is configured to have a partially flat surface upon which the retraction spring **638** rests, and a partially inclined surface such that the shoulder can be easily forced through the opening of the partition **634** by the firing spring **624**. The upper part of the shoulder **618** is configured to have a substantially flat surface to be captured by the opening of the partition **634** and prevent upward travel of the inserter rod **614** for retraction until released.

(78) The upper portion **602** comprises the first chamber **612** in which the firing spring **624** is captured. The firing spring **624** is disposed concentrically about the inserter rod **614** and is captured between an upper wall of the first chamber **612** and the cross-member **616** of the inserter rod **614**. In doing so, the firing spring **624** is configured to constantly urge the inserter rod **614** downward. Prior to use, the inserter rod **614** is held in an up position by one or more of the projections **610** of the hinged latch **606**. Specifically, an inner surface of the hinged latch **606** comprises one or more of the projections **610** which extend a slight distance from the inner surface of the hinged latch **606**,

and which block the travel of the cross-member **616** of the inserter rod **614**. In such a position, the firing spring **624** is compressed and the latches **606** capture the upper portion of the needle assembly as shown in FIG. **13***b*. The capture of the upper portion of the inserter rod **614** by the projections **610** of the latches **606**, and the force applied by the firing spring **624** while in the pre-use position, also serves to secure the upper housing **602** to the bottom housing **604** prior to use. Once the latches **606** are released from the inserter rod **614**, the upper housing **602** is free of the lower housing **604** and can be lifted away as shown in FIGS. **13***c*-**13***d*.
(79) The upper housing **602** can further comprise an opening **642** which can serve to support the firing spring **624** in position, and serve to guide the inserter rod **614** during use. The opening **642** can further reveal an extended portion of the inserter rod **614** as shown in FIG. **13***b* such that a user can confirm visually or by touch that the elements are all present and are properly assembled and ready for use.
(80) The lower portion **604** comprises a second, third and fourth chamber **626**, **628** and **630**. The second chamber **626** is substantially open at an upper surface to slidably receive the inserter rod **614** as guided by the cross-member **616** as urged downward by the firing spring **624** when released. The second chamber **626** comprises at least one projection **633** which extends inward from an inner surface of the second chamber **626**. In doing so, the projection **633** provides a downward travel limit of the inserter rod **614** through the contact between the cross-member **616** and the projection **633**. As noted above, the width of the second chamber and the cross-member **616** are configured such that the inserter rod **614** is centered and guided by each.
(81) The second and third chambers **626** and **628** are separated by a partition **634** having an opening **640** through which the inserter rod **614** extends. The opening **640** of the partition **634** is configured to have an inclined upper opening surface through which the inclined lower surface of the shoulder **618** can more easily pass as urged downward by the firing spring **624**. The lower surface of the opening **640** of the partition **634** is configured to be substantially flat such that the flat upper surface of the shoulder **618** cannot pass back though the partition **634** until released for retraction as described in greater detail below. Further, the partition **634** comprises at least one segment extending some distance from an outer surface of the lower element **604** (i.e., an extended user lever) such that the partition **634** can be deflected by the user for retraction as described in greater detail below.
(82) The third and fourth chambers are also separated by a partition **636**, which also includes an opening through which the inserter rod **614** extends. The third chamber **628** further comprises the retraction spring **638**. The retraction spring is positioned concentric with the inserter rod **614**, and is captured within the third chamber **628** between the partition **636** and the shoulder **618** of the inserter rod **614**. In doing so, the retraction spring **638** is configured to constantly urge the inserter rod **614** upward.
(83) Prior to use, the firing spring **624** in the upper portion **602** is compressed and the retraction spring **638** in the lower portion **604** is relaxed as shown in FIG. **13***b*. During use, the release of the firing spring **624** operates the inserter as described in greater detail below, and further serves to compress the retraction spring **638** as shown in FIG. **13***c*. After use, the inserter rod **614** is held in position by the contact between the shoulder **618** and the partition **634** as shown in FIG. **13***d*. To retract the inserter rod **614**, the user than presses on the extended portions of the partition **634**. As shown in FIGS. **13***d* and **13***e*, the restrictive opening **640** of the partition **634** serves to hold the inserter rod **614** in the down position. When the insertion is completed and the upper portion **602** is removed, the user can press the extended portions of the partition **634**, and the opening **640** is enlarged and allows the shoulder **618** to retract upward as urged by the retraction spring **638** as shown in FIG. **13***e*.
(84) While in the pre-use position, a large portion of the lower portion **604** remains open to the end of the device. In doing so, the set **642** can be positioned on the extended needle **622**, at an opposite side of the end **620** of the inserter rod **614**. The set **642** can be gently held within the portion **604**

through contact with the walls of the portion, and/or through contact with the inserter needle **622**. As noted elsewhere, the set can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the two-part inserter.

(85) As shown in FIGS. **13***a*-**13***f*, the compression of the latches **606** releases the projection latch on the needle assembly of the inserter rod **614**, permitting the firing spring **624** to drive the needle **622**, set **642**, and adhesive pad into the region of the skin beneath the portion **604**, and also compress the retraction spring **638**. That is, upon release, the inserter rod **614** is free to travel downward as urged by the trapped spring **624**. In doing so, the inserter rod **614**, including its end **620** and needle **622** travel downward through the portion **604**, urging the set **642** downward with it. At or before reaching the travel limit of the inserter rod **614**, the set is positioned, and the device can be removed as shown in FIG. **13***f*, thereby leaving the set **642** at the desired insertion site.

(86) Then, to retract the inserter rod **614**, including its end **620** and needle **622**, the user then presses on the extended portions of the partition **634**. As shown in FIGS. **13***d* and **13***e*, the restrictive opening **640** of the partition **634** serves to hold the inserter rod **614** in the down position. When the insertion is completed and the upper portion **602** is removed, the user can press the extended portions of the partition **634**, and the opening **640** is enlarged and allows the shoulder **618** to retract upward as urged by the retraction spring **638** as shown in FIG. **13***e*. The inserter rod **614** is prevented from completely exiting the lower portion **604** by the contact between the lower end **620** and the partition **636**.

(87) As noted above, the exemplary embodiment of the present invention illustrates an insertion device **600** that can be constructed in such a way as to separate high-cost parts from low-cost parts, keeping the former in a reusable mechanism while allowing the latter to be safely disposed as shown in FIG. **13***a*. A firing mechanism consisting of the hinged latches **606** and an extended, large spring is installed onto a disposable mechanism which includes the set **642**, the needle **622**, the adhesive pad, and a portion of the insertion mechanism. As shown, the firing spring **624** is compressed and the latches **606** capture the upper portion of the needle assembly or inserter rod **614** as shown in FIG. **13***b*. The device is now ready to be placed and fired.

(88) Once the user squeezes the latches **606** on the upper part of the device, the needle assembly or inserter rod **614** becomes free to move, and is driven downward by the firing spring **624**, piercing the skin, inserting and adhering the set **642** as shown in FIG. **13***c*. The downward motion of the needle assembly or inserter rod **614** also drives two secondary latches on the lower portion of the device outward to capture the inserter rod **614** within the opening **640** and compresses the return or retraction spring **638**. After removal of the reusable part **602** of the device **600**, the lower part of the device **604** remains in place and the needle will still be inserted in the skin as shown in FIG. **13***d*. To remove the needle **622**, the latches or the member **634** can be deflected as shown in FIG. **13***e* to allow the return or retraction spring **638** to extract the needle **622** and render the lower part **604** of the assembly **600** inert and disposable. The inserter rod **614** is prevented from completely exiting the lower portion **604** by the contact between the lower end **620** and the partition **636**.

(89) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective “squeeze-type” inserter as desired by a user. As shown by way of example, a “squeeze-type” inserter device **700** is shown in FIGS. **14***a*-**14***d*. As shown in FIGS. **14***a*-**14***d*, an exemplary embodiment of the present invention can be activated by pressing the device against the targeted portion of the skin and then squeezing a portion of the inserter body.

(90) An exemplary construction of the squeeze-type insertion device **700** can comprise a body having at least three portions **702**, **704** and **706**. The portions **702** and **706** can comprise rigid structures, having a flexible user-deflectable portion **704** disposed therebetween. The upper portion **702** can comprise a substantially cylindrical outer surface having a rounded and closed end, and an opposite end which seamlessly mates with the outer surface of the user-deflectable portion **704**. The lower portion **706** can have a larger diameter, substantially cylindrical outer surface, with a

contour configured to seamlessly mate with the outer surface of the user-deflectable portion **704**. Accordingly, the user-deflectable portion **704** can have a substantially cylindrical outer surface having a contour to provide seamless mating with the outer surface of the upper and lower portions **702** and **706**. The portions **702**, **704** and **706** can be constructed as a single element wherein each section is separated from the adjacent section by one or more cuts, which allow the segments to maintain assembly, but allow for the deflection of the user-deflectable portion **704** as described in greater detail below. Further, an inserter rod **712** is slidably captured within the device **700** and extends through each portion as guided by an inserter rod cross-member **714**. The inserter rod **712** further comprises a lower planar end **722** and an inserter needle **724**, which can be secured within a center opening of the inserter rod **708**, and can extend downward from the inserter rod **712** at the end **722**. Both the cross-member **714** and the lower end **722** are configured to have a width substantially equal to the width of the chamber in which each is positioned to facilitate alignment and travel of the inserter rod **712** during use.

(91) As shown in FIGS. **14***a*-**14***c*, the upper portion **702** provides a first chamber **708** in which a firing spring **710** is captured. Specifically, the firing spring **710** is positioned concentric with the inserter rod **712**, and is captured at one end by an upper wall of the first chamber **708**, and at an opposite end by the inserter rod cross-member **714**. In doing so, the firing spring **710** is configured to constantly urge the inserter rod **712** downward.

(92) Prior to use, the inserter rod **712** is held in position by one or more projections **716**. Specifically, an inner circumference of the user-deflectable portion **704** comprises one or more of the projections **716** which extend a slight distance from the inner circumference of the user-deflectable portion **704**, and which block the downward travel of the cross-member **714** of the inserter rod **712**. The projections **716** are provided at an uppermost point of the portion **704** as this point undergoes the greatest degree of deflection during user deflection as described in greater detail below. Accordingly, in such a position, the degree of deflection of the projections **716** is maximized as shown in FIG. **14***b* to simplify the release of the inserter rod **712**.

(93) The user-deflectable portion **704** provides a second chamber **718** through which the inserter rod **712** is positioned and which includes a travel limit element **720**. Although any part of the user-deflectable portion **704** can be compressed by a user, an exemplary embodiment of the present invention can provide one or more designations on an outer surface of the user-deflectable portion **704** to identify each as a preferred “squeeze button” area. Once the user-deflectable portion **704** is compressed by the user as shown in FIG. **14***b*, the projections **716** release the cross-member **714** of the inserter rod **712** and the firing spring **710** urges the inserter rod **712** downward through the second chamber **718** until the cross-member **714** is restricted from further downward travel by the travel limit element **720** as shown in FIG. **14***c*. As noted above, the portions **702**, **704** and **706** can be constructed as a single element wherein each section is separated from the adjacent section by one or more cuts, which allow the segments to maintain assembly but allow for the deflection of the user-deflectable portion **704**. Accordingly, the engagement between the portions **704** and **708** can be deflected in a manner similar to that of the deflection between the portions **702** and **704** described above in regard to the release of the inserter rod **712**.

(94) While in the pre-use position, a large portion of the lower portion **706** remains open to the end of the device. In doing so, a set **726** can be positioned on the extended needle **724**, at an opposite side of the end **722** of the inserter rod **712**. The set **726** can be gently held within the portion **706** through contact with the walls of the portion, and/or through contact with the inserter needle **724**. As noted elsewhere, the set **726** can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the squeeze-type inserter.

(95) As shown in FIGS. **14***a*-**14***d*, the compression of the user-deflectable portion **704** releases the projection latch **716** on the needle assembly of the inserter rod **712**, permitting the firing spring **710** to drive the needle **724**, set **726**, and adhesive pad into the region of the skin beneath the portion **706**. That is, upon release, the inserter rod **712** is free to travel downward as urged by the trapped

spring **710**. In doing so, the inserter rod **712**, including its end **722** and needle **724** travel downward through the portion **706**, urging the set **726** downward with it. At or before reaching the travel limit of the inserter rod **712**, the set **726** is positioned, and the device **700** can be removed as shown in FIG. **14***d*, thereby leaving the set **726** at the desired insertion site.

(96) Such an exemplary embodiment of the present invention provides a more convenient means of actuating an insertion device by simply using "squeeze" buttons to initiate the process as shown in FIG. **14***a*. As shown in FIG. **14***b*, buttons on the side of the device at the user-deflectable portion **704** can be pressed inward, unlatching the needle assembly or inserter rod **712** and permitting it to be propelled downward by the firing spring **710** as shown in FIGS. **14***b* and **14***c*. The spring-loaded needle delivers the set **726** and adhesive pad to the skin. The device **700** can then be removed as shown in FIG. **14***d*, thereby leaving the set **726** at the desired insertion site.

(97) In contrast to the other inserter devices described herein, the materials of the squeeze-type inserter **700** allow for the squeeze action of the one or more portions, or the construction of one or more portions having thinner, thereby deflectable parts.

(98) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective skin contact-type inserter as desired by a user. As shown by way of example, a "skin contact-type" inserter device **800** is shown in FIGS. **15***a*-**15***e*. As shown in FIGS. **15***a*-**15***e*, an exemplary embodiment of the present invention is shown wherein the insertion device can be activated by pressing the device up against the targeted portion of the skin.

(99) An exemplary construction of the insertion device **800** can comprise an upper and lower portion **802** and **804**, respectively. The upper portion **802** can be comprised of a dome-shaped element into which a slidably engaged lower portion **804** can be captured. Specifically, the upper portion **802** can comprise a substantially round, dome-shaped element having a flattened portion at an uppermost point. The upper portion **802** can further comprise one or more deflectable tabs **806**, or cut portions, about a body circumference as described in greater detail below. The inner surface of the tabs **806** are provided with an incline **824** and a projection **818** at the uppermost point of each incline.

(100) Within the upper portion **802**, exemplary embodiments of the present invention comprise an inserter rod **808** slidably disposed within the upper portion and which is held in place by the tabs **806**. As shown in FIGS. **15***b*-**15***e*, the inserter rod **808** comprises a planar end **814** wherein a notch **816** is proved at each side of the planar end **814**, which engages the projections **818** of the tabs **806** that extend into the upper portion **802**. Accordingly, prior to use, the projections **818** secure the notches **816** of the planar end **814** of the inserter rod **808**, preventing movement of the inserter rod **808**. The inserter rod **808** further comprises an inserter needle **828**, which can be secured within a center opening of the inserter rod **808**, and can extend downward from the rod at the planar end **814**.

(101) The upper portion **802** further comprises a firing spring **810** that is disposed concentrically with the inserter rod **808** and which is captured between the end **814** of the inserter rod **808** and one or more features of the upper flattened surface of the upper portion **802**. In the exemplary embodiment shown, the upper flattened surface of the upper portion **802** can comprise a molding **820** having a cup-shaped opening **812** that is sized to capture and hold an end of the firing spring **810**. In this position, the firing spring **810** serves to constantly urge the inserter rod **808** downward. However, as noted above, prior to use, the inserter rod **808** is held in place by the tabs **806**.

(102) Accordingly, the device **800** further comprises the slidably engaged lower portion **804** which is slidably captured at an end of the upper portion **802** by one or more contacting tabs **822** of portion **804**, and tabs **832** of the upper portion **802**. The lower portion **804** has a diameter slightly less than an inner diameter of the upper portion **802** such that the lower portion **804** is gently held in place prior to use but upon contact with a skin surface, can be easily slid upward into the upper portion **802**. In doing so, the tabs **822** of the lower portion **804** travel against an inner

circumference of the upper portion **802**, and onto the tabs **806**. Specifically, the tabs **822** of the lower portion **804** travel against the incline **824** of the tabs **806**, which forces the tabs **806** outward and releases the projections **818** from the notches **816** of the planar end **814** of the inserter rod **808** as shown in FIG. **15***c*. Once released in such a manner, the inserter rod **808** is free to travel downward as urged by the firing spring **810**.

(103) While in the pre-use position, a large portion of both the upper and lower portions **802** and **804** remain open to the end of the device **800**. In doing so, a set **826** can be positioned on the extended needle **828**, at an opposite side of the planar end **814** of the inserter rod **808**. The set **826** can be gently held within the upper portion **802** through contact with the walls of the upper or lower portion, and/or through contact with the inserter needle **828**. As noted elsewhere, the set **826** can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the skin contact inserter.

(104) As shown in FIGS. **15***b*-**15***d*, the slidable motion of the lower portion **804** into the upper portion **802** releases the pin latch **818** on the needle assembly of the inserter rod **808**, permitting the firing spring **810** to drive the needle **828**, set **826**, and adhesive pad into the region of the skin beneath the lower portion **804**. That is, upon release, the inserter rod **808** is free to travel downward as urged by the trapped firing spring **810**. In doing so, the inserter rod **810**, including its end **814** and needle **828** travel downward through the lower portion **804**, urging the set **826** downward with it. At or before reaching the travel limit of the inserter rod **808**, the set **826** is positioned, and the device **800** can be removed as shown in FIG. **15***e*, thereby leaving the set **826** at the desired insertion site.

(105) As described above, the lower portion **804** of the device becomes a movable actuation mechanism. In this case, a circumference or ring around the lower perimeter of the lower portion **804** of the device, surrounding the set **826** and needle **828**, is placed against the skin surface (not shown). The entirety of the device **800** is then pressed firmly downward. The lower ring of the lower portion **804** telescopes into the main body of the upper portion **802** of the device and as it travels inward/upward, it deflects the multiple latches around the perimeter provided by the tabs **806**, which permit the firing spring **810** to drive the needle **828**, set **826**, and adhesive pad into the skin. The device **800** can be removed as shown in FIG. **15***e*, thereby leaving the set **826** at the desired insertion site.

(106) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a needle handle and/or shroud in some manner as desired by a user. As shown by way of example, a needle handle and shroud **900** is shown in FIGS. **16***a*-**16***c*. In doing so, an enhancement can be provided to current manually inserted needles for set insertion by implementing a plastic handle **902** and a substantially circular and hinged shield **904** for use with the needle **906** and set (not shown). The handle **902** permits a sure grip on the needle **906** for insertion, while the circular shield **904**, as shown, helps to firmly press the set and set adhesive into place as shown in FIG. **16***a*. After placement, when the needle **906** is withdrawn, the hinged portions of the shield **904** can be folded down at each of hinges **908**, as shown in FIG. **16***b* and the cross-sectional view of FIG. **16***c*, to allow for safer needle disposal.

(107) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective skin pinch-type inserter as desired by a user. As shown by way of example, a “skin pinch-type” inserter device **925** is shown in FIGS. **17***a*-**17***d*. As shown in FIGS. **17***a*-**17***d*, an exemplary set can be packaged with a skin pinch-type inserter **925** with which a user can grasp a portion of the user's skin S (see, e.g., FIG. **17***a*) to improve insertion of the set. The exemplary insertion device **925** can be integrated into a mechanism as described in greater detail below that incorporates features to do this.

(108) An exemplary construction of the insertion device **925** can comprise a large clothespin-like clamp **926**, having opposite sides or legs **928** which rotate about a pin or other securing means **930**.

Such an exemplary clamp **926** can further provide a space therein in which a number of remaining components can be centered. The large clothespin-like clamp **926** can include the wide, manually-actuated legs **928** as shown in FIGS. **17***a*-**17***b*. The legs **928** can be configured to be at slight angles while at rest such that a width between the legs **928** at the skin contact surface is wider than a width between the legs **928** at opposite ends. When activated by a user, the user places the entire device **925** against the skin surface such that the wider opening contacts the skin surface. The user than grasps the legs **928** at a point near the skin surface and "pinches" the legs **928** into a substantially parallel position, such that the legs **928** extend vertically from the skin surface as shown in FIG. **17***c*. That is, the device can be placed on the skin surface, surrounding the target area, and the pinching legs **928** can be squeezed inwards. When pinched, the legs **928** in contact with the skin surface serve to prepare the skin surface in a number of different ways for set placement, including, but not limited to, stretching the skin at the site, leveling the skin at the site and/or raising the skin at the site.

(109) The squeezing of the legs **928** is also tied to a latching mechanism in the central section of the device **925**. That is, when a desired degree of skin pinch is achieved, the device automatically begins the process of releasing and placing the set. To do so, the device **925** can further comprise a first and second chamber therein. The first chamber **932** is secured to the second chamber **934**, which is secured at opposite corners to an inner surface of the legs **928**. The legs **928** further comprise at least one articulated pin **936** which is inserted into the first chamber **932** when the device is at rest as shown in FIG. **17***b*. The articulated pin **936** can include one or more joints or segments along its length, such that there is no interference with the movement of the legs **928** during operation.

(110) The articulated pin **936** extends from an inner surface of the legs **928**, through an opening in the side of the first chamber **932** and secures a planar end **938** of an inserter rod **940**. As shown in FIG. **17***b*, in doing so, the pin **936** holds the inserter rod **940** in an up and retracted position. As the legs **928** are pinched, the articulated pin **936** is pulled clear of the planar end **938** which allows the inserter rod **940** to move forward as urged by a firing spring **946**.

(111) As shown in FIGS. **17***b*-**17***c*, the firing spring **946** is captured in the second chamber **934**. Specifically, the first and second chambers are in communication via an opening therebetween through which the inserter rod **940** extends. The planar end **938** of the inserter rod **940** is captured in the first chamber, and can be held at an upper and retracted position in the first chamber by the articulated pin **936**. The remainder of the inserter rod extends into the second chamber **934** and terminates at an opposite planar end **944**, and further comprises an inserter needle **942**. The inserter needle **942** can be secured within a center opening of the inserter rod **940**, and can extend from the rod at the planar end **944** into the second chamber **934**. Both planar ends **938** and **944** are configured to have a width substantially equal to the width of the chamber in which each is positioned to facilitate alignment and travel of the inserter rod **940** during use.

(112) The firing spring **946** is placed concentrically about the inserter rod **940** and is captured by the inserter rod **940** within the second chamber **934**, between the end **944** and an upper wall of the second chamber **934**, and is configured to be in a compressed state prior to use, and upon release of the articulated pin **936**, is further configured to urge the inserter rod **940** downward toward the insertion site. As shown in FIG. **17***b*, while in the pre-use position, a large portion of the second chamber **934** at an opposite side of the planar end **944** remains open to the end of the device. In doing so, a set **948** can be positioned on the extended needle **942** within the second chamber **934**, at an opposite side of the planar end **944** of the inserter rod **940**. The set **948** can be gently held within the second chamber **934** through contact with the walls of the chamber, and/or through contact with the inserter needle **942**. As noted elsewhere, the set **948** can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the skin pinch inserter.

(113) As shown in FIGS. **17***b*-**17***c*, the motion of the legs **928** releases the pin **936** latch on the

needle assembly of the inserter rod **940**, permitting the firing spring **946** to drive the needle **942**, set **948**, and adhesive pad into the pinched region of the skin. That is, upon release of the articulated pin **936** from the first chamber **932**, the inserter rod **940** is free to travel downward as urged by the trapped firing spring **946**. In doing so, the inserter rod **940**, including its planar end **944** and needle **942** travel downward through the second chamber **934**, urging the set **948** downward with it. At or before reaching the travel limit of the inserter rod **940**, the set **948** is positioned, and the device **925** can be removed as shown in FIG. **17***d*, thereby leaving the set **948** at the desired insertion site.
(114) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective folding/retractable inserter as desired by a user. As shown by way of example, a "folding/retractable" inserter device **1000** is shown in FIGS. **18***a*-**18***e*. As shown in FIGS. **18***a*-**18***e*, an exemplary set can be packaged with a hinged, vertical protruding handle **1002** which improves the ability of the user to place it and insert it as shown in FIGS. **18***b*-**18***c*, but which can be folded out of the way as shown in FIGS. **18***a* and **18***d*. To do so, the inserter **1004** can be constructed having a hinge, pivot pin or pivot point, or other flexible-type element **1006** which allows the handle to extend from the set **1008** at a number of angles, especially when the set **1008** is adhered to a skin surface.
(115) As shown the cross-sectional views of FIGS. **18***b*-**18***d*, the folding/retractable inserter **1000** can comprise the handle **1002** in which a chamber **1010** is provided to contain an insertion needle **1012** and retraction spring **1014**. The handle **1002** further comprises at least one opening **1016** through which a user-accessible button **1018** can extend. As shown in FIG. **18***b*, the opening **1016** and user-accessible button **1018** restrict travel of an end of the needle **1012** such that the retraction spring **1014** is compressed and held in the state shown in FIG. **18***b*. In such a position, the inserter can be used in a conventional manner. Once the set is in place, the user can press the button **1018** such that the needle **1012** is released and the retraction spring **1014** can retract the needle **1012** from the set and into a protected and covered position within the chamber **1010** of the handle **1002** as shown in FIG. **18***c*.
(116) As more clearly shown in FIG. **18***c*, the handle **1002**, hinge **1006**, and chamber **1010** can further comprise an opening **1020** through which the needle **1012** can be extended and retracted. The opening **1020** is aligned when the handle **1002** is in the upright and perpendicular position. Once the needle **1012** is retracted into the chamber **1010**, the handle **1002** can be rotated about the hinge **1006** to any number of positions, but preferably is rotated to a down and substantially parallel position as shown in FIG. **18***d*. Once in this position, the inserter is left with the set at the site, but maintains a low profile due to the rotation of the handle **1002**. Further, no danger is presented by the needle **1012** of the inserter as it is fully retracted into the chamber **1010** of the handle **1002**, and is blocked from further escape as the opening **1020** is now blocked.
(117) As noted above, the device **1000** can include the user-accessible button **1018** at any convenient location along the handle **1002**, such that the button **1018** locks the needle **1012** in an extended position for insertion as shown in FIG. **18***b*. After insertion, the button **1018** can be activated or toggled, allowing the internal retraction spring **1014** to retract the needle **1012** vertically and automatically to a safe position within the chamber **1010** of the handle **1002** as shown in FIG. **18***c*. With the needle **1012** fully retracted into the handle body, the handle can be folded down by way of the hinge **1006** as shown in FIG. **18***d*. Such a hinge **1006** can further comprise a detented hinge to assist in maintaining the folded or upright position of the handle **1002** as shown in FIG. **18***e*, and to ensure that in the folded position, the sharp component is completely inaccessible and the device assumes and maintains a low profile. To do so, a number of detents **1022** can be provided to gently secure the handle **1002** in either the up or down positions.
(118) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective connection method between the tube connection and set in some manner as desired by a user. As shown by

way of example, a “squeeze-type” connection device **1100** is shown in FIGS. **19***a*-**19***c*. As shown in FIGS. **19***a*-**19***c*, a connection method is provided as a squeeze latch **1100** having one or more hinged and rotatable push buttons within the device that when pressed, release a shoulder, notch or other element of the set.

(119) For example, as shown in FIG. **19***a*, inside the body of the set **1104**, two push button latches **1102** are provided comprising rotatable L shaped elements, with pivot points **1108** located as shown in FIGS. **19***a* and **19***c*, and which can be used to secure the tube connection **1106** with the set **1104**. The latches **1102** can be urged into the securing position of FIG. **19***a* using springs, material resilience or other means (not shown). Squeezing the exposed portions of the two latches **1102** toward the center of the device **1100** causes the internal arms of the latches **1102** to swing free of the connection with the set **1104** and release it as shown in FIG. **19***c*. Once released, the tube connection **1106** and latches **1102** can be lifted from the set **1104**. In a similar manner, squeezing the exposed portions of the two latches **1102** toward the center of the device **1100** can be used to place the tube connection **1106** back on the set **1104**. Further, such a latching embodiment allows the tube connection **1106** to rotate about the set **1104** prior to, during and after assembly.

(120) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a connection tool and a set configured for use with such a connection tool in some manner as desired by a user. As shown by way of example, a connection tool **1154**, tubing connection **1152**, and set **1158** are shown in FIGS. **20***a*-**20***d*. That is, the device of FIGS. **20***a*-**20***d* illustrate an exemplary embodiment of a tool removable connection. Since user accessible connection mechanisms can be bulky or inadvertently actuated, a compromise can be to provide a release mechanism, but to require the use of a removal tool to trigger it as shown in FIG. **20***a*.

(121) As shown in FIGS. **20***a*-**20***c*, the tubing connection **1152** residing on top of the set **1158** will have a low profile, and the housing of the tubing connection **1152** comprises one or more small access openings **1162** as shown. While these openings **1162** are small enough to resist inadvertent disconnection, they permit insertion of specialized features on the external connection tool **1154**, which actuates the hinged and rotatable latches **1156** positioned inside the tubing connection **1152** to release it as shown in the cross-sectional views of FIGS. **20***b*-**20***d*.

(122) Specifically, the low profile tube connection **1152** is secured to the set **1158** using the hinged and rotatable latches **1156** positioned inside the tubing connection **1152**. The tube connection **1152** comprises one or more rotatable brackets **1156** which are configured to rotate about pins **1170**. Each of the rotatable latches comprises a shoulder **1172** configured to capture and securely hold a similar shoulder **1174** provided upon the set **1158**. Accordingly, in an assembled positions as shown in FIG. **20***b*, the brackets **1156** are rotated into a position that secures the set **1158** That is, when the connection tool **1154** is not assembled with the low profile tube connection **1152** and set **1158**, the hinged and rotatable latches **1156** are in upright and secure positions as shown in FIG. **20***b*. Any number of means can be used to maintain the latches **1156** in such an upright position such as springs, material elasticity, and/or contact with the skin surface. While in this position, the latches **1156** serve to capture and secure one or more of the shoulders **1174** of the set **1158**.

(123) The connection tool **1154** comprises a substantially cylindrical body having one or more projections or pins **1160** extending from a lower surface. The lower surface can be contoured to more closely match the upper surface of the tube connection **1152**. Further, the number and arrangement of pins **1160** and/or openings **1162** can be provided based upon the set such that a particular connection tool can be used with only one or more sets, or a universal tool can be provided to work with all sets. The tool **1154** can be made of any lightweight, strong material such as plastic or metal, and the pins **1160** can be constructed of similar materials to simplify manufacture.

(124) When the connection tool **1154** is pressed against an upper surface of the low profile tube connection **1152**, one or more of the pins **1160** extending from the tool **1154** enter the connection

1152 through one or more of the openings **1162** and contact the latches **1156**. A notch **1164** can be provided in the tool **1154** to allow clearance for the tube **1166**, and the tool can further comprise a contour **1168** and/or other surface or gripping features. Once engaged, the pins **1160** of the tool **1154** press the latches **1156** downward, from the upright and secured positions thereby releasing the set **1158** as shown in FIG. **20***c*. After the connection **1152** is released from the set **1158**, the tool **1154** and tube connection **1152** can be lifted free of the set **1158** as shown in FIG. **20***d*.
(125) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements **1200** to manage the tubing of the device in some manner as desired by a user. As shown by way of example, tube management elements **1204** and **1206** are shown in FIGS. **21***a*-**21***c*. As shown in FIGS. **21***a*-**21***b*, one or more loop control elements **1204** and **1206** can be provided to manage a length of tubing **1202** as desired by the user. Specifically, such tubing management can comprise one or more tubing pull ties **1204** and **1206**. The two ties **1204** and **1206** can be constructed having integral pull tabs **1210** and **1212**, respectively, and can be installed on a loop **1208** of tubing **1202** as shown. Pulling on the pump and set ends of the tubing **1202** causes additional slack of the loop **1208** to be released, and the loop of FIG. **21***b* to contract as shown in FIG. **21***a*. Pulling outward on the pull tabs **1210** and **1212** can then be used to cause the loop **1208** to expand, and to reduce slack in the tubing as shown in FIG. **21***b*.
(126) As shown in FIG. **21***c*, one or more of the pull ties can further comprise a clip **1214** to attach the loop of tubing to a user, and which can be pressed in the direction of arrow A to release the clip and loop of tubing from a user's belt or other clothing.
(127) Further, one or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to manage tubing connections of the device in some manner as desired by a user. As shown by way of example, a “breakaway” coupling and/or connecting engagement between tubes **1226** and **1240** is shown in FIGS. **22***a*-**22***c*. An advantage of the exemplary embodiments illustrated in FIGS. **22***a*-**22***c* is the provision of a secure, leak-free connection, that is easily assembled with a push connection action by the user, and which can disengage automatically when tubing interference occurs, without creating great degrees of tension. That is, the exemplary embodiments illustrated in FIGS. **22***a*-**22***c* preferably allow disengagement with minimal “pulling” of opposite tube ends such that the set or pump is not pulled free of the user.
(128) Each exemplary embodiment illustrated in FIGS. **22***a*-**22***c* preferably operates as a push connection. The tubing pieces can have male and female ends in which a selected feature provides the desired degree of engagement. As shown in FIG. **22***a*, the male end **1228** of tube **1226** can have a tapered shape, with a diameter and surface characteristics to serve as an in-line, sealable, press-fit with the female end **1234** of tube **1240**. In FIG. **22***b*, the male end **1230** can include one or more projecting detents **1242** to provide a snap-fit with similar recessed detents provided in the female end **1236**. In yet another exemplary embodiment, the male and female ends **1232** and **1238** can further include magnets **1244**, arranged to provide a magnetic attraction between tubes (i.e., north/south magnetic arrangements). In yet other embodiments, one of the magnets can be replaced with a metal element (not shown) which results in a substantially similar attraction with the remaining magnet.
(129) In each embodiment, the coupling and/or connecting engagement between tubes allows for easy disconnect and/or safety breakaway in the event of an accidental snag of the tubing. Further, any one of the exemplary embodiments can be further configured to provide an audible indication of disconnections, such as a “pop” sound or other alert. Although FIGS. **22***a*-**22***c* show coupling and/or connecting engagements between tubes **1226** and **1240**, in yet other embodiments of the present invention, the features can be used for connections between tube and set, tube and reservoir, and tube and pump.
(130) Still further, one or more of the exemplary embodiments of the present invention described

herein can be further provided with other additional features or elements to manage the tubing of the device in some manner as desired by a user. As shown by way of example, a tube management element **1250** is shown in FIGS. **23***a*-**23***b*. As shown in FIGS. **23***a*-**23***b*, an elastic accordion element **1254** is provided to manage a length of tubing **1252** as desired by the user. The length of the tubing **1252** can be controlled by a back-and-forth looped portion **1256** of the tubing **1252**, wrapped in the elastic binding **1254**, as shown. The elastic binding **1254** can be provided with one or more stitched or otherwise created seams **1258**, to hold and/or direct an entering and exiting segment of tubing to maintain proper operation of the elastic binding during operation. This elastic array of tubing can be expanded and contracted as necessary to allow for the ideal management of the tubing.

(131) Still further, one or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to minimize the tubing of the device in some manner as desired by a user. Great lengths of tubing can often result in a number of undesirable problems. Accordingly, where it is possible to do so, an exemplary embodiment of the present invention provides a system **1275** which can be safely implemented using a short tube **1278** as shown in FIGS. **24***a*-**24***c*.

(132) The short tube **1278** provided between the set **1276** and the pump **1280** can be minimal in length, including no slack that may pose a danger of tangling or fouling. The short tube **1278** can also be short enough to allow for disconnection at the pump **1280**, such as through any of the disconnecting means described above, leaving the short length of tubing still attached to the site. In an exemplary embodiment of the present invention, the short tube **1278** can be between 2 and 12 inches in length, and preferably between 3 and 9 inches in length and more preferably between 4 and 6 inches in length. The connection **1282** can be a quick-disconnect type, and can be configured to be released by pressing one or more buttons **1284** of the pump **1280** as indicated by the arrows A.

(133) Still further, one or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements **1285** to secure, contain, and/or conceal the tubing of the device in some manner as desired by a user. As shown by way of example, a tube containment pouch **1288** is shown in FIG. **25***a*. The kit can include the pouch **1288** that permits excess tubing **1290** to be bundled up and stowed in such a way as to allow only the required length **1286** to be freed for use. The pouch **1288** can be constructed in a manner substantially similar to that of fabric bag and having a Velcro enclosure, but is not limited thereto. For example, another exemplary embodiment is shown in FIG. **25***b* in which a flexible part **1292** can be provided that permits the tubing **1286** to be easily inserted and stored out of the way. The flexible part **1292** can be constructed of any suitable material, such as rubber, and can be provided with a number of slotted openings **1294** and **1296**, with varying slot **1298** dimensions to facilitate both tubing sizes, and ease of tubing insertion, as desired by the user. The part **1292** can be provided as a half-sphere to maximize both tube containment spacing while maintaining a manageable shape configuration. The pouch **1288** or containment part **1292** can be discarded with the tube.

(134) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to conceal, decorate, or alter the appearance of the device in some manner as desired by a user. As shown by way of example, an assembled device **1300** of FIG. **26***a* illustrates a number of such additional features or elements to conceal, decorate, or alter the appearance of the device. In the exemplary embodiment shown, the additional elements are not provided to conceal the device but rather, decorate or alter the appearance in a desired manner. Specifically, instead of providing means to conceal the set **1302** and tube **1304**, the inserted set is decorated or ornamented with customizable additional parts **1306** and **1308**. For example, as shown in the exploded view of FIGS. **26***b* and **26***c*, the set can be provided with an image element **1306** that can be secured to the set **1302** either with a snap fit or other friction-type,

adhesive-type, or other attachment, or can be trapped against the set **1302** by a cover piece **1308**. In such a manner, the set **1302** can be personalized with the image element **1306** trapped, protected, and/or further customized by the clear, snap-on cover piece **1308**.

(135) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to simplify gripping and user positioning of the inserter. Such an exemplary feature is shown in element **1325** of FIGS. **27***a*-**27***b*. In the exemplary embodiment shown in FIGS. **27***a*-**27***b*, an inserter **1326** can be provided with one or more finger loops for use in the placement of the device. In the embodiment shown, the finger loops **1328** allow for two-finger placement as illustrated in FIG. **27***b*. The set can be placed using such an insertion device **1326** with the two finger loops **1328** as shown, which enhance stability and precision of set placement. Although two completely encircled finger loops are shown, embodiments of the present invention are not limited thereto. In yet other embodiments of the present invention, a combination of complete and partial loops (i.e., spurs) can be used.

(136) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements for the use of site preparation. Such an exemplary feature is shown in element **1350** of FIG. **28**. In the exemplary embodiment shown in FIG. **28**, an inserter **1352**, such as those described above, can be provided with a spray mechanism **1354** at an upper most point of the inserter which contains a small amount of site preparation contents. The spray mechanism can be provided with a finger actuator **1356** that can be used by a user to spray the contents upon a site for site preparation. The contents can comprise, but are not limited to, anesthetic, disinfectant, or a combination of both, and can be applied by spraying from a small reservoir (not shown) and outlet incorporated into the insertion device.

(137) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements for the provision and use of an advanced design infusion set. Such an exemplary advanced design infusion set is shown comprising, in part, elements **1360** of FIGS. **29***a*-**29***c*. FIG. **29***a* is a cross-sectional view of the hub **1362** of the infusion set, in which the hub **1362** provides an annular fluid reservoir and/or fluid path **1364** which can significantly minimize the penetration distance of the tubeset connector into the hub while maintaining a sufficiently large tubeset connector geometry.

(138) The infusion set, as with others described herein, can be provided for the continuous subcutaneous infusion of insulin or other medications. The infusion set comprises the hub **1362**, which is affixed to a skin surface **1366** of a body with adhesive as described above, and further comprises a tubeset **1368** which creates and maintains a fluid path from an infusion pump and/or reservoir (not shown) and the hub **1362**. The exemplary hub **1362** illustrated in FIG. **29***a* comprises a polyurethane catheter **1370** extending from the hub **1362** through the skin surface **1366** and into the subcutaneous tissue. The hub can be constructed of materials having a viscoelastic property, and which are flexible, such as thermoplastic elastomer (TPE), thermoplastic urethane (TPU), silicone, or similar materials. As with many of the exemplary embodiments of the present invention described herein, the hub **1362** is configured to have a low profile to minimize interference, such as catching on clothing or fixed objects such as doorframes or cabinets. The hub **1362** can be further configured as described herein to have one or more clear or transparent features to enable site visualization and inspection.

(139) The central portion of the hub **1362** can comprise the annular space **1364** in fluid communication with the catheter **1370** lumen. The annular space **1364** can be configured in any number of ways to minimize deadspace within the hub **1362**, yet allow insertion needle penetration and subsequent closure of the penetration site, and provide tubeset needle penetration from any number of rotational positions of the tubeset as described in greater detail below. An advantage of providing such a hub **1362** with an annular interior is the ability to minimize the penetration distance of the tubeset connector into the hub while maintaining a sufficiently large tubeset connector geometry. Yet another advantage is the ability to minimize the volume of insulin or other

medication left in the hub fluid cavity.
(140) In an exemplary embodiment of the present invention, the annular space **1364** comprises a shallow or narrow central portion such that an upper surface, to be penetrated by the insertion needle **1372**, is close to the catheter **1370** lumen. A peripheral area of the annular space **1364** is provided with a wider space extending along the inner wall of the hub **1362**. In doing so, the engagement with the tubeset connector is simplified, and further allows tubeset engagement at any rotational position.
(141) The hub **1362** can be inserted through the skin surface **1366** and into the subcutaneous tissue via the insertion needle **1372** that extends through the top surface of the hub **1362**, through the annular space **1364** and through the catheter **1370** lumen. After insertion into the skin, the insertion needle **1372** (and needle hub) can be withdrawn and the resulting hole in the hub top surface can be configured to self-seal due to the elastomeric qualities of the hub materials after setting. That is, the annular fluid space **1364** can reseal any insertion openings generated by the insertion needle **1372**, and allow penetration by a tubeset connector needle as described in greater detail below.
(142) The exemplary tubeset connector **1368** also comprises, at least in part, the same or similar construction materials. The tubeset connector **1368** can be constructed of a viscoelastic material and utilizes the material characteristics to create and maintain a connection with the hub **1362**. The tubeset connector **1368** comprises an elasticized ring **1374** with an inner diameter, shape and profile that matches a central, recessed or grooved portion **1375** of the hub **1362**, and a tube **1378**. As shown in FIG. **29***a*, the central portion **1375** of the hub **1362** can comprise a groove or contour for guiding, receiving and securing the elasticized ring **1374**. To do so, the elasticized ring **1374** can be gently expanded by the user and placed into the central portion **1375** such that when released, the elasticized ring **1374** relaxes into a size and shape that secures the tubeset **1368** with the hub **1362**.
(143) At one location of the elasticized ring **1374** inner surface, the tubeset needle or spike **1376** is provided and protrudes radially inward. Accordingly, when the elasticized ring **1374** is placed onto the hub **1362** as described above, the tubeset needle **1376** punctures the wall of the central portion **1375** of the hub **1362** and thereby creates a fluid path from the tubeset **1368** to the annular cavity **1364** of the hub **1362**. The elastomeric qualities of the hub materials allow the needle punctures to self seal upon needle removal. In doing so, the tubeset **1368**, using the elasticized ring **1374** connector, can be affixed to the hub **1362** in any rotational alignment position. Further, the elasticity of the materials allows the elasticized ring **1374** to be removed from one position and repositioned elsewhere as desired, and each unused insertion site is sealed.
(144) One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements for the provision and use of an advanced design catheter. Such an exemplary advanced design catheter is shown in element **1380** of FIG. **30**. FIG. **30** is a cross-sectional view of a catheter including one or more cross-drilled holes and a splined lumen in accordance with an exemplary embodiment of the present invention. In the exemplary embodiment shown, the catheter **1380** can be constructed of a body temperature softening polyurethane or similar material, such as a proprietary Vialon material, and include one or more features including a splined lumen and holes or openings along a body length, including cross-drilled holes. For example, the catheter can comprise a splined lumen **1382** to prevent kinking, and flow occlusion in the event of a kinked catheter or otherwise. The splines **1382** can be provided as members raised from the inner wall of the catheter **1380** and/or provided as members depressed into the inner wall of the catheter **1380**, to be uniformly spaced and to extend along the axis of the catheter. For example, the splines can comprise a linear axis to run parallel to the axis of the catheter and/or can comprise a helical axis, either right-handed or left-handed, to form a helix curve about the axis of the catheter. Although four equally-spaced splines **1382** are shown in FIG. **30**, embodiments of the present invention are not limited thereto.
(145) Additionally, the catheter **1380** can comprise on or more holes or openings, such as the cross-

drilled holes **1384** shown in FIG. **30**. As shown, the holes can be aligned as resulting from drilling, punching or otherwise molding through holes at points along the body of the catheter. Any number of openings or holes can be provided, either uniformly over the body of the catheter or in arrangements of varying concentrations, and can comprise openings of approximately ¼ of the inner diameter of the catheter, but each are not limited thereto. The holes can be provided to facilitate infusion of the contents, such as insulin or other medication, in the subcutaneous tissue adjacent to the catheter in addition to, or instead of, the tissue at the catheter tip.
(146) To this point, a number advanced, improved, and novel new components, elements and packaging of current and future insulin infusion sets, have been described. Accordingly, a number of advantages and improvements over existing systems and methods include the feature of providing the set and some or all of its peripherals in one package. Many of the proposed insertion devices both insert and retract the needle and then cover it for safe disposal. The described fluid connections from the pump to the site require no moving parts and allow for easy disconnection. Other features allow the user to inspect the insertion area for irritation after insertion via a magnifying window in the site, and tube management is obtained by using a tube reel or other embodiment that allows the user to manage excessive tube length conveniently and adjust length as needed. The described insulin reservoir and tube are pre-filled and do not require priming by the user, and the concealment and decorative patches allow the user to cover and protect the site either inconspicuously or with distinction. The described site targeting ring allows the user to more accurately locate the site prior to insertion, and the disposable wipes and sprays combine disinfectant and anesthetic application steps into one step.
(147) Although only a few exemplary embodiments of the present invention have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention.

Claims

1. An infusion set, comprising: a set, comprising a skin contacting surface and a catheter extending from the set; a flexible ring having an integral fluid path therethrough, wherein the flexible ring is configured to expand to a first diameter to allow placement over the set, and contract to a second diameter to securely circumscribe an outer diameter of the set; a tubing length secured to the flexible ring; and a fitting or spike disposed at an end of the tubing length and extending into the flexible ring, the fitting or spike configured to sealably mate with an opening of the set to permit communication of a medicament therebetween.
2. The infusion set of claim 1, wherein the set comprises a hub constructed of a viscoelastic material, wherein the fitting comprises a cannula configured to sealably pierce the hub.
3. The infusion set of claim 2, wherein the viscoelastic material comprises at least one of a thermoplastic elastomer (TPE), thermoplastic urethane (TPU), or silicone.
4. The infusion set of claim 1, further comprising at least one tubing connection, the at least one tubing connection comprising at least one of a tapered tubing end, a tubing end having one or more detents, and a magnetically attractable tubing end.
5. The infusion set of claim 4, wherein the at least one tubing connection is configured to generate an audible indication when disconnected.
6. The infusion set of claim 1, further comprising a medicament supply, the medicament supply comprising at least one medicament container, a tubing segment and a tubing connection connectable to a pump that is connectable to an opposing end of the tubing length, wherein the medicament container is configured to indicate a content quantity therein.
7. The infusion set of claim 1, wherein the catheter comprises a splined lumen, wherein the splined

lumen comprises a helical axis.

8. The infusion set of claim 1, wherein the catheter comprises at least one hole or opening along a body length thereof.

9. The infusion set of claim 1, wherein the catheter is constructed of a body temperature softening polyurethane.

10. The infusion set of claim 1, wherein the set comprises at least one push button element, wherein operation of the at least one push button element is configured to release a tube connection from the set.

11. The infusion set of claim 1, wherein the catheter comprises a splined lumen, wherein the splined lumen comprises a linear axis.

12. The infusion set of claim 1, wherein the flexible ring is hollow.